# LEMON: Factor-Level Semantic Alignment for Graph–Text Pairs

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**Abstract.** Graph–text evaluation often misses a central failure mode: a sentence may preserve the entity names while changing or omitting the relation encoded by the graph edge. We introduce LEMON (Linguistically Enriched Measure of Ontological Normalization), a diagnostic metric for bidirectional knowledge-graph (KG)–text fidelity. LEMON decomposes each graph predicate into weighted semantic factors and evaluates both KG→Text coverage and Text→KG recoverability. Its goal is not to be the most severe scalar detector. Instead, it explains relation-level errors by localizing them to argument roles, relation class, direction, polarity, or textual evidence. On WebNLG, Forward LEMON reaches 0.792 and Reverse LEMON reaches 0.779. A MINE-STYLE comparison confirms that node information is easier to recover than edge information, while worked examples and controlled perturbations show why a factor trace is more informative than a single edge-failure label. The same inventory format is applied to DrugProt and BC5CDR, where biomedical predicates make role, direction, polarity, and causality explicit. The inventories and weights are auditable design resources, not yet expert-certified semantic gold standards. Code and reproducibility materials are available at `https://github.com/Cogniscy/lemon`.

**Keywords:** Graph-to-text evaluation · Semantic alignment · Knowledge graphs · WebNLG · Biomedical relation extraction · Predicate decomposition

## 1 Introduction

Although a graph edge is compact, it encodes more than the names of two entities. In an edge $(s, p, o)$, the predicate $p$ fixes the relation between the arguments: who did what, to whom, in which direction, and with which semantic commitment. A graph-to-text system can mention the entities and still lose the predicate. The reverse case is similar: a reconstructed graph is useful only if it recovers the relation, not merely the endpoints.

This distinction matters in ordinary examples. A sentence can mention *Alan Bean* and *Wheeler, Texas* without saying that Wheeler is his place of birth. A biomedical abstract can keep the same chemical and protein names while turning

inhibition into activation. Standard graph-to-text evaluation uses sentence-level metrics such as BLEU, METEOR, chrF++, TER, BERTScore, and BLEURT, as well as human criteria such as data coverage, relevance, correctness, and fluency [1–3]. These signals are useful, but they rarely identify which part of a predicate was preserved, weakened, or lost.

LEMON treats the predicate as the unit of diagnosis. Each predicate is decomposed into typed, weighted semantic factors. For example, *birthPlace* contains a person-like subject, a biographical relation, and a place-like object. A DrugProt inhibition predicate contains a chemical actor, a gene/protein target, negative regulation, directionality, and evidence form. The score evaluates whether these factors are expressed in text or recovered in a reconstructed graph, and it records which factors are missing.

The method is deliberately narrower than general semantic similarity. The source graph supplies the semantic structure, and LEMON checks whether the relevant predicate factors survive across graph and text. This makes the metric suitable for graph–text fidelity settings, including RDF-to-text evaluation and relation-centered biomedical transfer, but not for open-ended text–text similarity.

The closest comparison point in this paper is MINE-STYLE scoring, inspired by KGGen's Measure of Information in Nodes and Edges (MINE) [22]. A node/edge score is closer to graph fidelity than a sentence metric because it separates recovered entities from recovered relations. Its limitation in our setting is resolution: missing relation class, reversed roles, and wrong polarity all collapse into a similar edge-level failure. LEMON keeps the graph-aware view but opens the edge into weighted semantic factors.

The experiments follow this argument. We first show that entity overlap on WebNLG hides relation-level errors. We then evaluate the same factor inventory in the reverse Text→KG direction, stress metrics with controlled perturbations, transfer the inventory format to DrugProt and BC5CDR, and run a small three-judge LLM probe to test factor-level recoverability.

The contributions are: (i) a bidirectional factor-level metric for KG–text fidelity; (ii) fixed predicate-factor inventories for WebNLG, DrugProt, and BC5CDR; (iii) reproducible forward, reverse, perturbation, component, and LLM-judge analyses; and (iv) a direct comparison with coarse node/edge and triple-style views. The paper is diagnostic rather than leaderboard-oriented: it shows not only that a relation failed, but whether the failure concerns role assignment, direction, polarity, relation class, or evidence.

## 2 Related Work

### 2.1 RDF-to-text benchmarks and sentence metrics

The WebNLG Challenge established RDF-to-text generation as a task in which systems verbalize DBpedia-derived triples [1]. WebNLG+ extended the task to bilingual and bidirectional generation [2], and GEM promoted broader evaluation

practice for generation systems [3]. Recent large language model (LLM) work on graph-to-text generation, including PlanGTG, has renewed attention to planning and attribution in graph-conditioned generation [4].

Reference metrics used in this area measure different forms of sentence similarity. BLEU and METEOR reward n-gram or alignment overlap; chrF++ compares character and word n-grams; TER measures edit operations; BERTScore and BLEURT use contextual or learned semantic matching [5–10]. These metrics are useful for fluency and reference similarity. They are not designed to say whether a source predicate, its argument roles, direction, polarity, or evidence was preserved.

### 2.2 Source-aware factuality and graph metrics

Source-aware metrics evaluate generated text against the input data. PARENT weights n-gram precision and recall by evidence in the source table and reference text [11]. Data-QuestEval uses question answering for data-to-text coverage without requiring a reference [12]. AlignScore learns an alignment function for factual consistency [13]. FactSpotter is the closest prior graph-to-text evaluation approach: it decides whether a triple $(s, p, o)$ is present in a generated text and correlates with human judgments for data correctness, coverage, and relevance [14]. LEMON complements this line by replacing a binary whole-triple decision with a weighted predicate-factor diagnosis.

Atomic factuality metrics take the opposite route. FActScore and VERISCORE decompose generated text into verifiable claims and check support against evidence [15, 16]. Semantic graph metrics such as Smatch, $S^2$match, and Smatch++ compare graph structures under node alignment, mainly for AMR-style parser evaluation [17–19]. LEMON is source-driven: the graph predicate is decomposed first, and the text is evaluated against that decomposition.

### 2.3 Text-to-KG extraction, MINE, and LLM judging

Text-to-KG evaluation asks whether an extractor retains useful information from plain text. KGGen introduces a language-model-based generator and MINE, a Measure of Information in Nodes and Edges, for testing node and edge retention [22]. Our MINE-STYLE baseline adapts this node/edge split to WebNLG graph–text pairs. It is not a reproduction of the full KGGen benchmark; it is a local baseline for asking whether a reconstructed graph retains nodes and relations. The limitation relevant here is diagnostic resolution: several semantic errors can become the same edge-level failure.

LLM-based evaluation is useful when exact lexical rules miss recoverable meaning, but it must be reported separately from rule-based scoring. G-Eval and MT-Bench show that LLM judges can scale evaluation while introducing preference and calibration risks [26, 27]. Related retrieval work also uses LLMs to resolve ambiguous relevance judgments, for example by filtering hard negatives when raw similarity scores are insufficient [28]. We use LLMs only in an

exploratory factor-judging probe; the main tables remain reproducible without API calls.

### 2.4 Behavioral metric calibration

A metric should also respond predictably to controlled changes. CheckList introduced behavioral tests for NLP models, including invariance and directional expectation tests [23]. Perturbation CheckLists adapt this idea to NLG metrics [24], and synthetic stress tests expose blind spots of model-based NLG evaluation [25]. Our perturbations follow this logic: relation deletion, argument swap, polarity flip, and relation blur test whether a metric reacts to intended semantic damage.

## 3 Method

### 3.1 Problem setting

Let a graph–text pair be $(G, T)$, where $G$ is a set of directed edges $e = (s, p, o)$ and $T$ is a verbalization. The forward task evaluates whether the text covers the information in the graph. The reverse task evaluates whether a graph reconstructed from the text, denoted $\hat{G}(T)$, preserves the information in the original graph. LEMON uses the same predicate-level representation in both directions.

### 3.2 Predicate factors and weights

Each predicate $p$ is mapped to a weighted decomposition:

$$D(p) = \{(f_i, r_i, w_i)\}_{i=1}^{n}, \quad \sum_i w_i = 1,$$

where $f_i$ is a semantic factor, $r_i$ is its role, and $w_i > 0$ is a salience weight. Roles cover predicate meaning, subject and object domains, value domains, and modifiers such as direction, polarity, causality, or evidence. The representation is coarser than a full ontology but richer than a single predicate label. A text that keeps both entities but loses the relation cue can therefore receive partial credit rather than full coverage.

The inventories are fixed before scoring. They are manually specified diagnostic resources built from predicate labels, argument roles, domain constraints, entity types, and lexical evidence cues. The weights are normalized salience priors. Predicate-meaning factors usually receive the largest share; domain factors and modifiers receive smaller shares when they are part of the predicate meaning. These weights are auditable design choices, not learned parameters or expert-certified semantic constants. An expert validation protocol has been prepared, but expert results are not included in this version.

Figure 1 shows two worked examples. MINE-STYLE collapses both cases into a coarse relation mismatch. LEMON preserves partial credit for the factors that remain correct and localizes the error to polarity or relation class.

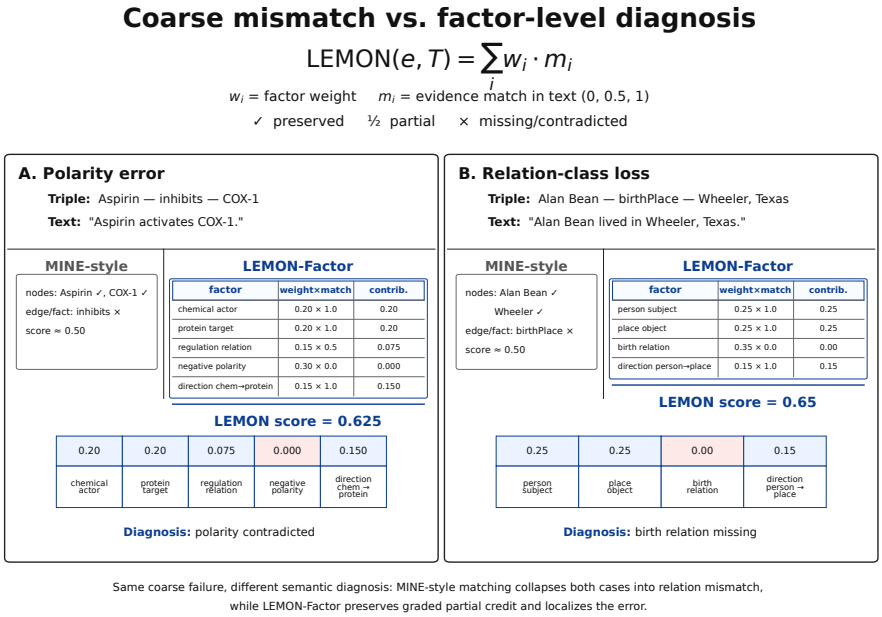


**Fig. 1.** Worked examples of factor-level scoring. The displayed weights come from the current predicate inventories and are normalized to sum to one. The MINE-STYLE view collapses both cases into a coarse relation mismatch, whereas LEMON assigns partial credit to preserved factors and localizes the failure to polarity or relation class. The examples illustrate scoring mechanics, not a separate benchmark.

### 3.3 Forward LEMON: KG→Text coverage

For an edge $e$, text $T$, and component $(f_i, r_i, w_i)$, an evidence function $E(f_i, r_i, e, T) \in [0, 1]$ estimates whether the component is expressed. In the main reproducible experiments, $E$ is implemented with fixed lexical and domain rules: entity and domain factors use normalized label or token evidence, while relation factors use predicate cues and factor cues. The edge-level forward score is:

$$S_F(e, T) = \sum_i w_i E(f_i, r_i, e, T),$$

because the weights are normalized. Example-level and corpus-level scores are averages over edges with available decompositions. The implementation stores covered and missing factors for each edge, so the scalar score is paired with a diagnostic trace.

### 3.4 Reverse LEMON: Text→KG recoverability

The reverse direction first constructs a rule-based reconstructed graph $\hat{G}(T)$ from the text using entity-label evidence and predicate lexical cues. Reverse

LEMON then compares each gold edge with its best recovered counterpart. Subject and object recovery contribute through node/domain evidence; predicate recovery contributes through overlap between the gold predicate's factors and the recovered edge evidence. The reverse score evaluates whether the source predicate's semantic factors survive text-to-graph reconstruction.

The current reconstructed graph is a lexical pseudo-extractor, not a full KG extraction model. This choice keeps the experiment reproducible and keeps the focus on metric behavior rather than extractor quality. Future work can replace $\hat{G}(T)$ with KGGen, GraphRAG-style extraction, OpenIE, or domain-specific extractors.

### 3.5 MINE-style node/edge baseline

To compare factor-level recoverability with node/edge information retention, we implement a MINE-STYLE baseline inspired by KGGen's MINE. Each gold triple is treated as a fact, and the reconstructed graph is searched for matching or related nodes and edges. The score separates node information, edge information, and a composite partial-credit node/edge score. We report it as a local baseline rather than a reproduction of the original benchmark.

### 3.6 Controlled perturbation calibration

A graph–text metric should react to meaning-destroying perturbations and remain comparatively stable under meaning-preserving perturbations. We create auditable text and graph perturbations while keeping the source graph fixed. Destructive operations include relation phrase deletion, object swaps, predicate swaps, node drops, and edge drops. Preserving operations include alias-like rewrites and template paraphrases. Weak-template cases are marked when a template preserves entity strings but may alter phrasing in a way that is less reliable. The perturbation files and operation manifests are saved with the repository.

## 4 Data and Experimental Resources

### 4.1 WebNLG sample

The WebNLG experiments use a stratified English sample built from RDF-to-text examples. The train split contains 200 examples and the development split contains 100 examples; both cover 16 WebNLG categories. The development set contains 308 graph edges. The sample is small by design: it keeps the complete scoring pipeline auditable while still exposing relation-level errors that are hidden by entity overlap.

### 4.2 Biomedical transfer resources

The biomedical transfer checks use two relation extraction corpora converted into the same graph–text JSONL format. DrugProt is a BioCreative VII resource for chemical–gene/protein relations with 13 relation types and 24,526 manually annotated relations [20]. In the current processed subset, the train split contains 500 abstracts, 3588 edges, 12 observed predicate types, and an average text length of 1729.5 characters. BC5CDR is a BioCreative V corpus of 1500 PubMed articles with chemical and disease annotations and 3116 chemical–disease interactions [21]. The converted BC5CDR train and development splits each contain 500 documents, with 985 and 961 CID edges respectively. These resources are used as transfer diagnostics only; we do not train or claim a new biomedical relation extraction system.

### 4.3 Predicate-factor inventories

The experiments use fixed JSON predicate-factor inventories. Each inventory contains a controlled factor vocabulary and one weighted decomposition per predicate. WebNLG uses 121 predicate decompositions and covers all 119 predicate labels observed in the evaluation files. DrugProt contains 13 biomedical relation decompositions and covers the 12 relation types observed in the processed subset. BC5CDR is intentionally a single-relation setting: its inventory decomposes the document-level chemical-induced disease relation into chemical exposure, disease outcome, causality, document-level evidence, and chemical-to-disease direction. This makes BC5CDR useful for testing whether the same factor format transfers to a biomedical relation with causality and document-level evidence, not for testing predicate-label diversity.

The inventories are rule-based diagnostic resources. Earlier LLM-generated decomposition candidates are kept as development material, but they are not treated as expert gold and are not used as the main evidence in the scoring tables below. The validator checks that coverage is complete for the current evaluation files and that each predicate decomposition has normalized weights.

## 5 Experiments

The experiments test one representation from several angles. WebNLG provides an explicit RDF-to-text setting. DrugProt and BC5CDR test whether the same factor format remains meaningful when predicates encode biomedical roles, direction, polarity, and causality. Main scoring uses JSONL inputs and fixed JSON factor inventories. LLM calls appear only in the reliability probe; they are not used to construct the rule-based tables.

### 5.1 Forward and reverse factor scoring

The first WebNLG experiment scores development texts using predicate decompositions and lexical cues. We run an error-driven resource expansion: missing

predicates are identified, rule-based decompositions are added, lexical cues are expanded, and coverage is recomputed. This separates missing evaluation resources from missing information in the text.

To test bidirectionality, we reconstruct a graph from each WebNLG text with a lexical pseudo-extractor and compute Reverse LEMON. The reverse score is compared with expanded Forward LEMON. A small forward–reverse gap would indicate that the same factor schema can support both KG→Text coverage and Text→KG recoverability.

### 5.2 Node/edge retention and perturbations

We compare Reverse LEMON with MINE-STYLE, which reports node information, edge information, and a composite node/edge score. A locked 50-fact subset shows how binary fact recoverability differs from partial-credit node/edge retention. These values are diagnostic baselines, not a reproduction of the full KGGen/MINE benchmark.

Controlled perturbations are injected into text and reconstructed graphs. The source graph remains fixed, so the test measures whether a metric reacts to intended semantic damage in the text. The final scoring uses five destructive operators: node deletion, relation-cue deletion, argument swap, polarity flip, and relation blur. Polarity flips are most informative for DrugProt and BC5CDR, where relation labels explicitly encode activation, inhibition, regulation, or causality.

We score the shared perturbation files with entity recall, label match, binary triple match, a MINE-STYLE node/edge harmonic score, a label-only LEMON variant, and the full LEMON score based on weighted predicate factors. The same scoring layer is used for the component analysis. An offline character n-gram vector baseline is reported as a lexical/topical control, not as a dense embedding benchmark.

### 5.3 Exploratory LLM factor judging

The LLM reliability probe evaluates semantic recoverability. We sample perturbed instances from WebNLG, DrugProt, and BC5CDR and ask three independent LLM judges to label each semantic factor as covered, partially covered, or absent. The judges are Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B, called through OpenRouter with JSON-schema constrained outputs. This probe instantiates the evidence decision with LLMs to test stability, but it is not human validation and it is not used in the main rule-based scores.

## 6 Results

### 6.1 From entity overlap to relation factors

The first WebNLG experiment shows why predicate resources must be explicit. With the initial factor dictionary, only 140 of 308 development edges could be

scored, and the missing decomposition rate was 0.546. After rule-based decomposition and cue expansion, all 308 edges had predicate factors and Forward LEMON rose from 0.368 to 0.792. This increase does not mean that all WebNLG texts are faithful; it shows that relation-level diagnosis requires predicate resources before text quality can be evaluated.

Lexical controls identify the added signal. Exact entity-label overlap is already high at 0.922 and does not change under predicate-resource expansion. Token cosine and Jaccard remain fixed because the graph–text strings do not change. LEMON changes because it checks relation roles, predicate meanings, and lexical evidence instead of counting entity mentions alone.

The reverse run checks the same representation from text back to graph. A lexical pseudo-extractor recovered 390 of 408 nodes and 196 of 308 edges from the WebNLG texts. Reverse LEMON reached 0.779, close to the expanded Forward score of 0.792. This does not evaluate a general-purpose extractor; it shows that one predicate-factor schema can support both KG→Text coverage and Text→KG recoverability.

The MINE-STYLE comparison exposes the coarse-grained difficulty. Its node/edge composite is 0.803, but this aggregate is dominated by node recovery. On the locked 50-fact subset, node information is 0.960, while edge information and rule-based fact recoverability are both 0.540. Entity mentions are easier to preserve and recover than the relations that connect them.

A relation-deletion sanity check makes this visible. Removing relation evidence changes exact entity-label coverage by only 0.002, while Forward LEMON drops by 0.211. Predicate cue coverage and MINE fact recoverability fall by 0.364 and 0.354. Thus, entity preservation is not relation preservation; relation-aware scores are needed to detect this failure mode.

### 6.2 Cross-domain perturbation sensitivity

The cross-domain perturbation files test which metrics react when controlled edits damage relation meaning in WebNLG, DrugProt, and BC5CDR. Table 1 summarizes the semantic layers that the current factor schema can audit. A perturbation can be diagnosed only when the corresponding layer is present in the inventory.

Figure 2 restates the perturbation comparison as an annotated matrix. The scale is linear and the cell values are explicit. The figure shows the intended distinction: MINE-style and triple-style scores are more severe scalar detectors, while LEMON provides a factor trace that remains tied to roles, polarity, direction, and evidence.

Table 2 reports the same sensitivity values as percentages of the normalized score range. The columns are different preservation signals, so the table should not be read as a global ranking. It shows how much each signal reacts to each controlled edit.

Entity recall drops least because entity strings often remain when relation meaning is damaged. Triple match and MINE-STYLE react more strongly, but they collapse the error into one edge/fact decision. LEMON is useful when the

**Table 1.** Semantic layers covered by the current factor schema. The rows are diagnostic layers, not independent models; each row identifies a kind of predicate information that can be preserved or damaged.

| Layer | What the layer tests | Scope and diagnostic use |
|---|---|---|
| Entity/domain | Subject/object mention and coarse type support; stressed by node deletion. | Separates entity survival from relation preservation. |
| Predicate cue | Lexical or factor evidence for the relation; stressed by relation cue deletion and relation blur. | Detects when entities remain but the relation is no longer expressed. |
| Role/argument | Which participant fills which predicate role; stressed by argument swap. | Identifies role reversals that entity-level scores may miss. |
| Direction | Directed effect or source–target orientation. | Applies when the inventory encodes directional predicates, mainly in biomedical relations. |
| Polarity | Activation, inhibition, causality, or negative regulation; stressed by polarity flip. | Applies only when polarity is explicit in the predicate inventory. |
| Evidence | Residual textual support for a source factor. | Separates intended perturbation damage from remaining contextual recoverability. |

**Table 2.** Perturbation sensitivity as normalized score loss (%). Entity recall checks source entity mentions; Triple match checks whole SPO recovery; MINE-STYLE is the adapted node/edge baseline; LEMON label-only removes factor decomposition; LEMON is the full weighted factor score.

| | Entity | Triple | MINE-style | LEMON | |
|---|---|---|---|---|---|
| Perturbation | recall | match | node/edge | label-only | LEMON |
| Node deletion | 32.6 | 67.5 | 65.7 | 32.3 | 35.2 |
| Edge deletion | 21.5 | 65.5 | 63.7 | 48.2 | 28.1 |
| Argument swap | 21.4 | 54.0 | 52.1 | 32.1 | 41.3 |
| Polarity flip | 21.5 | 54.6 | 52.6 | 33.7 | 41.4 |
| Relation blur | 21.6 | 56.8 | 54.9 | 38.5 | 28.1 |

reason for failure matters: it reacts to argument swaps and polarity flips when those dimensions are encoded, and it stays quiet when the inventory does not contain that dimension.

Table 3 is an internal component analysis, not a comparison with MINE-STYLE or triple matching. Removing participant-role factors produces the largest positive gain for the full model over the ablated variant, so role factors are the largest observed source of diagnostic sensitivity. Evidence, polarity, and direction have smaller effects, mainly in biomedical predicates where these factors are explicit. Label-only scoring can produce a stronger scalar reaction when surface labels disappear, but it cannot say whether the damage concerns roles, direction, polarity, or evidence.

### 6.3 LLM factor judging as a complementary view

The LLM probe evaluates semantic recoverability rather than rule-based perturbation labels. Each judge saw the source edge, the factor inventory, and the perturbed text, then labeled each factor as covered, partial, or absent. Table 4 summarizes Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B.

The mean LLM factor score is 0.809, pairwise agreement is 0.597, and agreement with rule-based perturbation labels is 0.561. These values indicate mod-

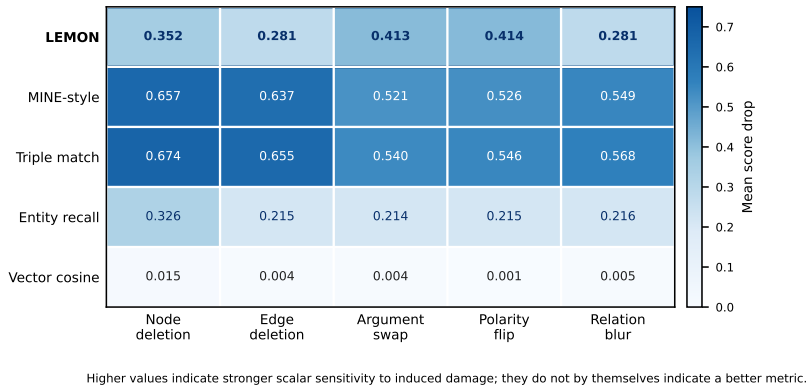


**Fig. 2.** Diagnostic perturbation matrix under controlled edits. Cell values are mean score drops, not absolute accuracy. Higher values indicate stronger sensitivity to induced damage. MINE-style and triple-match are coarse edge/fact detectors; vector cosine is the offline character n-gram vector baseline. LEMON is less severe in aggregate, but its response is linked to explicit predicate factors.

erate stability rather than expert validation. Rule-based perturbations record intended damage; LLM judges estimate how much of the original factor remains recoverable from context. The two views are complementary: LEMON exposes controlled relation-level failures, while model-based judging shows which failures remain inferable.

## 7 Error Analysis

Two error classes recur. The first is inventory incompleteness. Predicates without decompositions, or with weak lexical cues, cannot be scored reliably. The error-driven expansion step exposes this problem before it contaminates the metric. After expansion, all 308 development edges had decompositions, but fallback decompositions remain less informative than expert-designed ones.

The second class is relation recoverability. Node information is high because WebNLG verbalizations often repeat entity labels, while edge information is substantially lower. The same pattern appears in the reverse experiment, in the MINE-STYLE node/edge gap, and in the relation-deletion perturbation. The difficult part of graph–text alignment is often the relation that links the entities, not the entity names themselves.

The factor trace is the practical output of LEMON. A low score can be traced to a predicate, a factor role, a missing cue, or weak subject/object evidence. Instead of returning only an aggregate number, the metric gives a debugging signal for datasets, generators, and extraction pipelines.

**Table 3.** Component sensitivity analysis for LEMON. Loss is normalized score loss under perturbation, reported in percent. Gain is the loss of full LEMON minus the loss after removing a component; positive gain means that the removed component contributed to sensitivity. This table explains LEMON internals rather than ranking LEMON against other metrics.

| Variant | Loss (%) | Gain vs. full (pp) |
|---|---|---|
| LEMON | 32.0 | 0.0 |
| Label-only | 41.7 | -9.7 |
| Unweighted | 31.0 | 1.0 |
| Minus roles | 18.5 | 13.5 |
| Minus direction | 31.0 | 1.0 |
| Minus polarity | 30.4 | 1.6 |
| Minus evidence | 29.0 | 3.0 |

**Table 4.** Exploratory LLM factor-judging probe. Judges are Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B. Mean factor score maps covered, partial, and absent to 1, 0.5, and 0; agreements are computed only on overlapping factor judgments.

| Probe | Items | Judges | Mean score | Pairwise agr. | Rule-label agr. |
|---|---|---|---|---|---|
| LLM factor judging | 45 | 3 | 0.809 | 0.597 | 0.561 |

## 8 Limitations

The study does not yet include expert human validation. The factor inventories are rule-based resources built from predicate labels, schema constraints, domain roles, lexical cues, and development-time candidates. The LLM probe adds a semantic recoverability check, but LLM judgments are not a replacement for expert annotation. Prior LLM-as-judge work shows that LLM judges can scale evaluation while introducing preference, calibration, and reasoning biases; the next version should add expert factor annotations and inter-annotator agreement [26, 27].

The evidence layer is limited. It can find labels and known cues, but it can miss implicit relations, long-range entailment, and paraphrases outside the cue dictionary. A stronger implementation should add embedding, entailment, or QA-based evidence while preserving the factor trace.

The reverse pipeline uses a lexical pseudo-extractor rather than a full text-to-KG system. This makes the experiment controlled, but the reverse scores should be read as metric-behavior diagnostics. Similarly, the MINE-STYLE baseline adapts KGGen's node/edge information idea to graph–text pairs; it is not a reproduction of the original MINE benchmark.

The scope is narrower than general semantic similarity. LEMON assumes that a source graph, relation annotation, or predicate inventory is available. It is therefore a graph–text fidelity metric, not a general-purpose text–text similarity metric. The current paper also does not evaluate dense embedding baselines for arbitrary sentence pairs and does not claim to replace embedding-based retrieval or semantic textual similarity systems.

The present inventories cover relation-level roles, direction, polarity, causality, and evidence where those dimensions are encoded by the predicate resources. They do not model pragmatic phenomena such as irony, implicature, presupposition, stance, or rhetorical force. Finally, the perturbations are auditable but template-like. They support construct-validity checks for relation deletion, argument swap, and polarity flip, but they do not prove robustness to natural paraphrase or state-of-the-art biomedical extraction performance.

## 9 Conclusion

LEMON evaluates KG–text fidelity by opening graph predicates into weighted semantic factors. The metric does not aim to replace sentence-level metrics, triple matching, or MINE-STYLE node/edge scoring. Its role is diagnostic: when a relation fails, it identifies whether the failure concerns relation class, roles, direction, polarity, or evidence.

The experiments support this distinction. On WebNLG, the same factor representation supports forward coverage and reverse recoverability. The MINE-STYLE comparison shows that nodes are easier to recover than edges, and the relation-deletion check shows that entity-label coverage can remain almost unchanged while relation coverage drops. Across WebNLG, DrugProt, and BC5CDR, perturbations show that LEMON reacts to the semantic dimensions represented in the inventory.

The component and LLM analyses clarify the result. Role factors contribute most to diagnostic sensitivity under controlled damage. Label-only scoring can be more severe as a scalar detector, but it cannot explain whether the damage concerns role assignment, direction, polarity, or evidence. LLM judges provide a complementary view of residual recoverability; they do not replace expert validation.

The main contribution is therefore a layer between whole-triple factuality and sentence-level similarity: LEMON gives a reproducible factor trace for graph–text errors that coarser scores collapse into one number. Future work should validate the predicate inventories with domain experts, replace the lexical evidence layer with stronger paraphrase and entailment signals, and repeat the reverse experiment with stronger text-to-KG extractors. Broader layers such as modality, quantification, presupposition, and pragmatic effects require separate annotation and remain outside the current results.

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